

Exploiting Delivery-Window Data to Classify Compliance Types and Estimate LATE

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Abstract

In randomized experiments with imperfect compliance, the local average treatment effect (LATE) requires observing treatment receipt, and the Wald estimator requires the monotonicity assumption that rules out defiers. I consider settings where receipt is unobserved but the experiment has a pre-treatment time series and a distinct treatment delivery window. I show that a two-sided break test applied to the delivery window can classify treated units into compliers, never-takers, and defiers, yielding three things the standard setup does not provide: an inferred compliance rate with closed-form bias correction, an empirical test of monotonicity, and a characterization of the complier subpopulation that can be projected onto the control group. I derive the delta method variance for the corrected LATE, exploiting the independence between the compliance classifier and the outcome that arises from the temporal structure of the experiment. In simulations, the corrected LATE is unbiased and the 95% confidence interval achieves coverage between 0.947 and 0.974 across twelve parameter configurations.

Keywords: LATE, compliance, defiers, monotonicity, instrumental variables, structural break

JEL codes: C21, C26, C12

1 Setting

Consider a randomized experiment with N units. Let $Z_i \in \{0, 1\}$ denote treatment assignment and $D_i \in \{0, 1\}$ denote actual treatment receipt, which is unobserved. The experiment proceeds in three temporal phases: a pre-treatment period with T_{pre} outcome observations per unit, a treatment delivery window with T_{treat} observations during which

the experimenter attempts to apply the treatment, and a post-treatment outcome period with T_{post} observations from which the outcome Y_i is constructed.

Following Angrist et al. (1996), each unit has a compliance type determined by the pair $(D_i(1), D_i(0))$: compliers $(D_i(1) = 1, D_i(0) = 0)$, never-takers $(D_i(1) = 0, D_i(0) = 0)$, always-takers $(D_i(1) = 1, D_i(0) = 1)$, and defiers $(D_i(1) = 0, D_i(0) = 1)$. In one-sided noncompliance, $D_i(0) = 0$ for all i , which rules out always-takers. The standard LATE framework additionally assumes monotonicity, ruling out defiers, so that the population consists of compliers (share π) and never-takers (share $1 - \pi$). The LATE is then identified by the Wald estimator:

$$\tau = \frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[D_i | Z_i = 1] - E[D_i | Z_i = 0]} = \frac{\text{ITT}}{\pi} \quad (1)$$

The problem addressed here is that D_i is unobserved, so π cannot be estimated directly. The temporal structure of the experiment provides a way around this, and, as a byproduct, an empirical check on the monotonicity assumption.

2 Compliance Classification via Break Tests

The treatment, when successfully delivered, produces a deterministic shift in the unit's outcome during the delivery window. I exploit this by comparing each treated unit's delivery-window outcome to its pre-treatment baseline. The test statistic for unit i with $Z_i = 1$ is:

$$t_i = \frac{\bar{y}_{i,\text{treat}} - \bar{y}_{i,\text{pre}}}{\hat{\sigma}_i \sqrt{1/T_{\text{pre}} + 1/T_{\text{treat}}}} \quad (2)$$

where $\hat{\sigma}_i$ is the pre-treatment standard deviation of unit i 's daily outcome.

A one-sided test ($\hat{D}_i = \mathbf{1}(p_i < \alpha)$) suffices to classify compliers and never-takers when monotonicity holds. But the temporal data supports a richer classification. A two-sided test at level α partitions treated units into three groups:

$$\hat{C}_i = +1 \quad (\text{complier}) \quad \text{if } t_i > t_{1-\alpha/2} \quad (3)$$

$$\hat{C}_i = -1 \quad (\text{defier}) \quad \text{if } t_i < -t_{1-\alpha/2} \quad (4)$$

$$\hat{C}_i = 0 \quad (\text{never-taker}) \quad \text{if } |t_i| \leq t_{1-\alpha/2} \quad (5)$$

For control units, $\hat{C}_i = 0$ by construction (one-sided noncompliance implies $D_i = 0$ regardless of type).

2.1 Testing Monotonicity

Define $\hat{\delta} = N_{\text{treat}}^{-1} \sum_{i:Z_i=1} \mathbf{1}(\hat{C}_i = -1)$, the share of treated units classified as defiers. Under monotonicity ($\delta = 0$), the expected share of false defier classifications is $\alpha/2$, the lower-tail false positive rate. A test of $H_0 : \delta = 0$ against $H_1 : \delta > 0$ can be conducted by comparing $\hat{\delta}$ to $\alpha/2$:

$$z_{\text{mono}} = \frac{\hat{\delta} - \alpha/2}{\sqrt{(\alpha/2)(1 - \alpha/2)/N_{\text{treat}}}} \quad (6)$$

Rejection provides evidence against monotonicity. If $\hat{\delta} \leq \alpha/2$, the data are consistent with monotonicity and all apparent defiers can be attributed to the lower tail of the null distribution.

Remark 1. *In the standard LATE framework, monotonicity is untestable without additional structure (Angrist et al., 1996). The delivery-window data makes it testable because defiers produce a distinctive signature: a significant negative break during a period when the treatment should produce a positive shift. This is specific to settings where the direction of the mechanical effect is known.*

2.2 Inferred Compliance Rate

Define the inferred complier share as $\hat{\pi} = N_{\text{treat}}^{-1} \sum_{i:Z_i=1} \mathbf{1}(\hat{C}_i = +1)$. Under the two-sided test, the false positive rate for classifying a never-taker as a complier is $\alpha/2$ (the upper tail).

Assumption 1. *The mechanical dose during the delivery window is sufficiently large relative to organic variation that $\text{TPR} = 1$ for compliers and the probability of a defier being classified as a complier is zero.*

Under Assumption 1 and monotonicity ($\delta = 0$):

$$E[\hat{\pi}] = \pi + (1 - \pi) \frac{\alpha}{2} \quad (7)$$

3 Corrected Estimator

Inverting gives a corrected compliance rate:

$$\pi_c = \frac{\hat{\pi} - \alpha/2}{1 - \alpha/2} \quad (8)$$

and a corrected LATE:

$$\hat{\tau}_c = \frac{\widehat{\text{ITT}}}{\pi_c} = \frac{(1 - \alpha/2) \cdot \widehat{\text{ITT}}}{\hat{\pi} - \alpha/2} \quad (9)$$

Proposition 1. *Under Assumption 1 and monotonicity, the corrected estimator $\hat{\tau}_c$ is consistent for τ . The bias ratio of the uncorrected Wald estimator $\widehat{\text{ITT}}/\hat{\pi}$ relative to the true LATE is $\pi/[\pi + (1 - \pi)\alpha/2]$.*

For $\pi = 0.75$ and $\alpha = 0.05$, the bias ratio is $0.75/0.7625 = 0.984$.

Remark 2. *When the one-sided test is used instead (monotonicity assumed a priori), the false positive rate is α rather than $\alpha/2$, and the correction becomes $\pi_c = (\hat{\pi} - \alpha)/(1 - \alpha)$.*

4 Inference

The corrected LATE is $\hat{\tau}_c = f(\widehat{\text{ITT}}, \hat{\pi})$ where $f(a, b) = (1 - \alpha/2) \cdot a / (b - \alpha/2)$. The delta method gives:

$$\text{Var}(\hat{\tau}_c) \approx \left(\frac{1 - \alpha/2}{\hat{\pi} - \alpha/2} \right)^2 \text{Var}(\widehat{\text{ITT}}) + \left(\frac{(1 - \alpha/2) \cdot \widehat{\text{ITT}}}{(\hat{\pi} - \alpha/2)^2} \right)^2 \text{Var}(\hat{\pi}) \quad (10)$$

where $\text{Var}(\widehat{\text{ITT}})$ is the standard difference-in-means variance and $\text{Var}(\hat{\pi}) = \hat{\pi}(1 - \hat{\pi})/N_{\text{treat}}$.

The two terms are independent because $\widehat{\text{ITT}}$ is computed from post-treatment outcome data and $\hat{\pi}$ from delivery-window data. This independence arises from the three-window temporal structure, which serves the same function as cross-fitting in the test-and-select estimator of Hazard and Löwe (2023) and the general DML framework of Chernozhukov et al. (2018), but as a structural partition of the data-generating process rather than a random partition of the sample.

The 95% confidence interval is $\hat{\tau}_c \pm 1.96 \cdot \hat{\text{se}}$.

5 Partial Identification When Assumption 1 Fails

When $\text{TPR} < 1$, the corrected compliance rate π_c overestimates π and the uncorrected estimator $\widehat{\text{ITT}}/\hat{\pi}$ understates the LATE. Together these bound the true LATE:

$$\frac{\widehat{\text{ITT}}}{\hat{\pi}} \leq \tau \leq \frac{(1 - \alpha/2) \cdot \widehat{\text{ITT}}}{\hat{\pi} - \alpha/2} \quad (11)$$

6 Monte Carlo Evidence

I simulate experiments with $N = 500$ (250 treated, 250 control), $T_{\text{pre}} = 30$, $T_{\text{treat}} = 6$, and $T_{\text{post}} = 30$. Baseline daily outcomes are drawn from $N(5, 3)$ across units with within-unit daily noise $\sigma = 2$. The mechanical dose is 17 per day during the delivery window. The causal effect is τ additional units per day for compliers during the post-treatment period, so the true LATE is $\tau \times T_{\text{post}}$. Each scenario uses 1,000 Monte Carlo replications. The one-sided test with correction $(\hat{\pi} - \alpha)/(1 - \alpha)$ is used for the main simulations (monotonicity is imposed in the DGP); the two-sided classification is exercised in the defier simulations below.

Table 1: Corrected LATE: bias and 95% CI coverage across parameter configurations.

Scenario	True LATE	Mean $\hat{\tau}_c$	Bias	Coverage	CI Width
Baseline ($\pi=.75, \tau=3$)	90	89.4	-0.6	0.974	42.6
$\pi=.50$	90	89.5	-0.5	0.972	65.9
$\pi=.90$	90	89.9	-0.1	0.959	33.7
$\tau=1$	30	29.6	-0.4	0.956	39.0
$\tau=10$	300	300.1	0.1	1.000	73.1
$\tau=0$ (null)	0	-0.3	-0.3	0.947	38.4
$N=100$	90	89.7	-0.3	0.974	96.9
$N=2000$	90	89.7	-0.3	0.969	21.3
$\sigma=5$	90	89.6	-0.4	0.969	39.5
Mechanical=5/day	90	90.0	0.0	0.967	42.8
$T_{\text{pre}}=7$	90	89.7	-0.3	0.963	42.6
$\alpha=.01$	90	90.0	0.0	0.974	42.7

Table 1 reports the results. Coverage lies between 0.947 and 0.974 in all configurations. Coverage holds under the null ($\tau = 0$, coverage 0.947). CI width scales with $N^{-1/2}$ and with π^{-1} . At baseline, the inferred compliance rate is $\hat{\pi} = 0.761$, the corrected rate is $\pi_c = 0.748$, and the true rate is $\pi = 0.750$.

6.1 Defier Detection

I add defiers to the DGP: a fraction δ of treated units experience a negative mechanical shift of the same magnitude during the delivery window (their outcome decreases by 17/day), and a negative treatment effect of $-\tau$ per day in the post-treatment period. The two-sided classification from equations (3)–(5) is applied.

Table 2 reports results from simulations with $\pi = 0.75$ and varying δ (the never-taker share adjusts to $1 - \pi - \delta$). When $\delta = 0$, the false defier rate is 0.005, well below $\alpha/2 =$

Table 2: Defier detection: monotonicity test and compliance classification.

True δ	$\hat{\delta}$	$\alpha/2$	Reject $H_0: \delta=0$	$\hat{\pi}_{\text{complier}}$	$\hat{\pi}_{\text{defier}}$
0.00	0.005	0.025	0.000	0.755	0.005
0.05	0.048	0.025	0.892	0.754	0.048
0.10	0.095	0.025	1.000	0.752	0.095
0.15	0.139	0.025	1.000	0.751	0.139

0.025, and the monotonicity test never rejects—the test is conservative under the null. At $\delta = 0.05$ (5% defiers), power is 0.892. At $\delta \geq 0.10$, the test rejects in every simulation.

7 Identifying Compliers in the Control Group

The classification in Section 2 operates only on treated units. However, compliance type is a pre-treatment characteristic of the unit, determined by how it would respond to treatment. If compliance type is predictable from pre-treatment observables—the trajectory of outcomes before the experiment, or other baseline covariates—then the classifier can be extended to control units.

Among treated units, the delivery-window test reveals compliance type. A model trained on these units to predict $\hat{C}_i$ from pre-treatment features X_i can then be applied to control units to obtain $\hat{C}_i(1)$, a prediction of what compliance type the control unit would have had under treatment. This is the principal score approach of [Frangakis and Rubin \(2002\)](#), where the principal score $P(D_i(1) = 1 \mid X_i)$ is typically estimated under parametric assumptions. The temporal structure here allows the principal score to be estimated nonparametrically from the treated group, where compliance is revealed by the break test, using standard supervised learning methods.

With predicted compliance types for both arms, the LATE can be estimated directly as a difference in means within the predicted-complier subpopulation:

$$\hat{\tau}_{\text{match}} = E[Y_i \mid \hat{C}_i(1) = 1, Z_i = 1] - E[Y_i \mid \hat{C}_i(1) = 1, Z_i = 0] \quad (12)$$

rather than as a ratio. The identifying assumption is that compliance type is independent of potential outcomes conditional on X_i , a form of unconfoundedness within principal strata. This assumption is strong but testable in the treatment arm: if pre-treatment features predict compliance well among treated units, the matching has content; if they do not, the prediction is uninformative and one falls back to the Wald estimator without loss. I leave the formal development of this extension to future work.

8 Discussion

The temporal structure of experiments with a pre-treatment series, a distinct delivery window, and a post-treatment outcome window provides more than a compliance rate. It provides a unit-level compliance classification that enables three things the standard LATE setup does not: a corrected Wald estimator with closed-form bias adjustment and valid inference (Sections 3–5), an empirical test of monotonicity via defier detection (Section 2.1), and a path to identifying complier-type units in the control group (Section 7).

The approach connects to several recent threads. [Hazard and Löwe \(2023\)](#) propose a test-and-select LATE estimator using cross-fitting for valid post-selection inference. [Abadie et al. \(2024\)](#) show that exploiting first-stage heterogeneity can substantially reduce the MSE of IV estimators. [Chernozhukov et al. \(2018\)](#) provide the general DML framework for two-step procedures. Principal stratification ([Frangakis and Rubin, 2002](#)) handles latent compliance types under parametric assumptions. The key requirement throughout is that the treatment produces a mechanical shift during the delivery window that is large relative to organic variation. When this holds, the delivery window functions as a revealed-compliance device, analogous to the placebo design of [Nickerson \(2005\)](#) in which control-group units are contacted with an unrelated message to reveal contactability. The temporal version requires no placebo arm, only that treatment delivery is concentrated in a known window separate from the outcome measurement period.

References

- Abadie, A., Gu, J., and Shen, S. (2024). Instrumental variable estimation with first-stage heterogeneity. *Journal of Econometrics*, 240(2):105425.
- Angrist, J. D., Imbens, G. W., and Rubin, D. B. (1996). Identification of causal effects using instrumental variables. *Journal of the American Statistical Association*, 91(434):444–455.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.
- Frangakis, C. E. and Rubin, D. B. (2002). Principal stratification in causal inference. *Biometrics*, 58(1):21–29.
- Hazard, Y. and Löwe, S. (2023). Improving LATE estimation in experiments with imperfect compliance. Working paper, Collegio Carlo Alberto.

Nickerson, D. W. (2005). Scalable protocols offer efficient design for field experiments.
Political Analysis, 13(3):233–252.